Value Stream Mapping Report

Process Name: Software Development Lifecycle

Customer: Internal Business Units

Supplier: Development Team

Product: Custom Software Application

Analysis Date: 2025-07-31

Analyst: Demo System

Key Metrics Summary

Total Cycle Time:	3840.0 minutes
Total Lead Time:	15600.0 minutes
Value-Added Time:	3360.0 minutes
Process Efficiency:	87.5%
Takt Time:	2400.0 minutes
Number of Processes:	7

Executive Summary

This value stream analysis examines the Software Development Lifecycle process serving customer Internal Business Units. The analysis identified 7 key process steps with a total cycle time of 3840.0 minutes and lead time of 15600.0 minutes. The current process efficiency is 87.5%, indicating significant opportunities for improvement through waste elimination and process optimization.

Waste Analysis Summary:
Total identified waste: 21000.0 minutes
Primary waste sources: motion, waiting, overprocessing, defects, underutilized_talent
Improvement potential: 134.6% lead time reduction

Process Details

Process	Cycle Time	Lead Time	Uptime	Operators	Efficiency	Value Added
Requirements Analysis	480.0 min	1440.0 min	90%	2	33.3%	Yes
Design & Architecture	720.0 min	2880.0 min	85%	3	25.0%	Yes
Development	1440.0 min	7200.0 min	80%	5	20.0%	Yes
Code Review	240.0 min	960.0 min	95%	2	25.0%	Yes
Testing	480.0 min	1920.0 min	88%	3	25.0%	Yes
Documentation	360.0 min	720.0 min	92%	1	50.0%	No
Deployment	120.0 min	480.0 min	75%	2	25.0%	No

Waste Analysis

Waste Type	Impact (minutes)	Description
Waiting	11400.0	Idle time waiting for materials, information, or equipment
Defects	7200.0	Effort to find and correct defects
Overprocessing	1680.0	More work than required by customer
Motion	360.0	Unnecessary movement of people
Underutilized Talent	360.0	Not fully utilizing employee skills and ideas

Improvement Recommendations

Recommendation 1: Waste Elimination

Priority: Medium
Description: Reduce motion waste
Action: Unnecessary movement of people
Expected Impact: Potential 360.0 min reduction in lead time

Recommendation 2: Waste Elimination

Priority: Medium
Description: Reduce waiting waste
Action: Idle time waiting for materials, information, or equipment
Expected Impact: Potential 11400.0 min reduction in lead time

Recommendation 3: Waste Elimination

Priority: Medium
Description: Reduce overprocessing waste
Action: More work than required by customer
Expected Impact: Potential 1680.0 min reduction in lead time

Recommendation 4: Waste Elimination

Priority: Medium
Description: Reduce defects waste
Action: Effort to find and correct defects
Expected Impact: Potential 7200.0 min reduction in lead time

Recommendation 5: Waste Elimination

Priority: Medium
Description: Reduce underutilized_talent waste
Action: Not fully utilizing employee skills and ideas
Expected Impact: Potential 360.0 min reduction in lead time

Recommendation 6: Equipment Reliability

Priority: Medium
Description: Improve equipment uptime
Action: Implement TPM (Total Productive Maintenance)
Expected Impact: Increase overall capacity and reduce variability